

ESTHERVILLE, IA, USA

WMO: 726499

Lat: 43.401N

Lon: 94.747W

Elev: 1317

StdP: 14.01

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 94971

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.2	-7.1	-17.3	2.2	-10.6	-13.0	2.8	-6.3	34.6	17.5	32.2	16.7	10.1	310	0.652

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	89.8	74.5	87.0	72.7	84.0	71.0	77.8	85.4	75.7	83.3	73.9	81.2	15.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.3	139.2	82.5	73.1	129.3	80.7	71.7	123.1	79.0	42.2	85.3	40.2	83.2	38.3	80.8	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
30.0	26.7	24.2		-18.4	94.7	5.7	2.8	-22.6	96.7	-25.9	98.4	-29.1	99.9	-33.3	101.9
				-18.6	81.4	5.5	1.8	-22.6	82.6	-25.8	83.7	-28.9	84.6	-33.0	85.9
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.8	16.0	19.8	32.6	46.4	58.5	68.4	72.0	68.6	62.2	48.6	34.4	21.0
	DBStd	22.01	12.97	12.85	12.45	10.02	8.94	6.57	5.67	5.38	8.12	9.66	10.99	11.96
	HDD50	4186	1053	846	553	181	26	0	0	0	7	143	478	899
	HDD65	7608	1518	1266	1004	561	241	39	10	26	147	515	917	1364
	CDD50	2667	0	0	14	73	288	551	681	576	373	100	11	0
	CDD65	615	0	0	0	3	39	140	227	136	63	7	0	0
	CDH74	5766	0	0	5	73	476	1339	2081	1067	617	106	2	0
	CDH80	1736	0	0	0	19	152	442	679	254	165	25	0	0
	WSAvg	11.6	13.1	13.2	13.1	14.0	13.3	10.9	8.5	7.6	9.3	11.2	12.6	12.7
	PrecAvg	28.0	0.6	0.6	1.7	3.0	3.7	4.4	3.6	3.5	3.1	2.0	1.3	0.6
(15) Precipitation	PrecMax	45.0	3.0	3.4	4.1	7.7	9.3	13.5	10.5	8.4	9.6	8.3	5.4	2.0
	PrecMin	18.5	0.0	0.0	0.0	0.6	0.9	1.4	0.3	0.2	0.2	0.1	0.0	0.0
	PrecStd	6.1	0.7	0.7	1.1	1.7	2.0	2.5	2.2	1.9	2.2	1.7	1.1	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.5	54.5	72.0	82.2	90.3	93.0	92.8	90.0	89.6	83.6	70.2	53.9
		MCWB	39.5	48.1	57.5	61.8	68.2	73.1	77.9	77.3	71.9	64.2	55.4	49.0
	2%	DB	39.1	46.1	63.1	74.2	84.0	88.8	89.7	85.8	84.3	75.6	62.5	46.0
		MCWB	35.3	42.3	53.5	57.7	65.1	71.9	76.9	75.0	69.9	61.1	52.8	41.0
	5%	DB	36.0	39.5	56.5	69.0	79.4	85.2	86.8	82.4	81.2	70.3	55.6	39.3
		MCWB	33.6	36.8	49.8	54.3	62.7	69.2	75.0	72.8	67.5	58.9	47.5	35.6
	10%	DB	33.1	36.3	49.9	63.5	74.4	81.7	83.8	80.4	77.2	64.7	50.3	36.0
		MCWB	31.6	34.2	45.3	52.4	60.7	67.6	72.9	71.1	65.7	55.6	44.6	33.8
	0.4%	WB	40.7	49.5	60.9	63.4	72.3	77.1	81.6	80.2	75.3	69.5	58.4	50.8
		MCDB	46.5	55.2	65.8	74.8	84.1	87.6	88.3	87.0	84.0	78.7	66.5	53.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.0	42.2	55.2	59.9	68.5	74.7	78.6	76.8	72.7	64.3	53.5	41.1
		MCDB	38.8	46.3	61.5	70.5	79.1	84.4	86.1	83.3	80.5	71.1	60.5	45.8
	5%	WB	33.7	37.3	50.4	56.9	65.8	72.5	76.8	74.6	70.5	60.6	48.9	36.8
		MCDB	35.6	39.7	56.0	66.5	76.2	81.5	84.3	80.8	77.7	68.1	55.1	39.6
	10%	WB	31.5	34.3	44.9	53.5	63.0	70.2	74.8	72.4	68.1	56.5	44.7	33.6
		MCDB	33.0	36.0	50.2	61.2	70.5	78.7	82.1	78.1	74.6	64.6	50.2	35.5
(35) Mean Daily Temperature Range	5% DB	MDBR	16.2	16.5	16.9	21.4	21.3	20.2	20.0	19.7	23.1	21.8	18.4	15.3
		MCDBR	19.1	18.7	27.0	31.9	28.4	24.6	21.5	21.5	26.5	29.5	27.6	21.3
		MCWBR	15.9	15.2	17.4	17.4	13.6	11.1	11.5	12.4	13.2	16.4	17.7	16.9
	5% WB	MCDBR	17.5	17.7	25.0	27.0	22.9	20.4	19.6	19.4	21.9	24.3	25.0	19.5
		MCWBR	15.0	14.4	16.9	16.4	12.6	11.4	11.8	12.3	12.6	15.7	17.3	15.9
(40) Clear-Sky Solar Irradiance	taub		0.271	0.282	0.310	0.360	0.383	0.401	0.416	0.407	0.365	0.327	0.286	0.266
	taud		2.435	2.420	2.446	2.336	2.343	2.344	2.283	2.292	2.381	2.480	2.521	2.475
	Ebn at Noon		278	293	297	287	282	276	270	269	273	272	270	269
	Edh at Noon		23	28	31	38	39	39	41	39	33	26	21	20
(44) All-Sky Solar Radiation	RadAvg		587	926	1212	1493	1750	1937	2067	1719	1385	924	620	457
	RadStd		62	99	118	138	123	146	90	122	91	120	48	42

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	-1.87	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (56 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page